

The role and evolution of knowledge network-based intellectual capital in the corporate university

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Abstract

Purpose – The purpose of this study is to explore the knowledge network-based intellectual capital of corporate universities and its co-evolution process with knowledge management activities.

Design/methodology/approach – An exploratory and multiple case study was conducted, investigating three Chinese corporate universities. Grounded theory was used for data analysis.

Findings – This paper finds that the intellectual capital of corporate universities comprises teacher network capital, knowledge process capital and knowledge ba. The steering wheel model is established through the synergistic interaction and promotion among these three types of intellectual capital. The interaction between intellectual capital and the knowledge network within corporate universities constructs the intellectual capital network, which plays the roles of coordinator, knowledge gatekeeper and innovation bridge. The intellectual capital of corporate universities is characterized by sequential inertia. Moreover, the intellectual capital and intellectual capital network are aligned with knowledge management activities at each stage of corporate university development, interacting and following the principles of ladder evolution.

Originality/value – A significant contribution of this paper lies in applying the concept of intellectual capital within the fourth-stage ecosystem to a broader range of knowledge networks. By exploring the dynamics and network of intellectual capital in corporate universities, especially the role of intellectual capital networks and the synergy between intellectual capital and knowledge management activities, this study enriches the existing research on knowledge management and intellectual capital of corporate universities. Furthermore, it advances the development of knowledge management promotion in corporate universities from a new perspective of intellectual capital.

Keywords Corporate university, Role and evolution, Intellectual capital, Knowledge management, Knowledge network

Paper type Research paper

1. Introduction

Corporate universities were initially established in the GM School of Design and Management and have since expanded rapidly in developed regions. According to statistics, approximately 80% of Fortune Global 500 companies have established their own corporate universities (Chen *et al.*, 2019). Meister (1997) provided the first definition of a corporate university as an education and training institution for lifelong learning aimed at improving employee performance. As the role of corporate universities has evolved, they have transformed into platforms for enterprise knowledge management (Scarso *et al.*, 2017; Chen and Xu, 2022). However, some corporate universities currently face challenges such as unrecognizable internal knowledge gaps, ineffective knowledge connections and limited knowledge incubation capabilities, which hinder their ability to promote knowledge innovation. Simultaneously, certain successful corporate universities have transformed into knowledge centers of enterprises by adopting effective



methods and utilizing available resources. These phenomena have prompted scholars and entrepreneurs to investigate the factors that influence the realization and enhancement of the knowledge management function of corporate universities.

The research perspectives that influence the effectiveness of knowledge management in corporate universities are diverse. Previous studies have examined the key factors that promote knowledge enhancement of corporate universities from the perspectives of human resources (Patrucco *et al.*, 2017), strategic management (Petr and Elena, 2018) and the resource-based view (Sletli and Grønhaug, 2017). Scholars generally believe that intellectual capital, which refers to knowledge that contributes to the value creation of enterprises, is a crucial factor in gaining competitive advantages (Yüksel *et al.*, 2022). The management and rational utilization of intellectual capital typically determine the success of enterprises (Chatterjee and Chaudhuri, 2022). An increasing number of scholars have conducted theoretical and practical research on intellectual capital and achieved remarkable results (Secundo *et al.*, 2018a; Fiano *et al.*, 2020). However, few scholars have explored the influencing factors that promote the knowledge management capabilities of corporate universities from the perspective of intellectual capital. Corporate universities serve as collections of multifactorial resources (Chen *et al.*, 2022), while intellectual capital encompasses a multi-dimensional concept consisting of human capital, structural capital, social capital and relational capital (Subramaniam and Youndt, 2005; Kianto *et al.*, 2017). Nevertheless, no scholars have yet studied the significant role played by intellectual capital in the knowledge management of corporate universities.

Bontis (1998) initially drew attention to the network attribute of intellectual capital but did not conduct an in-depth analysis of the network relationship among various structural elements. Existing studies have demonstrated the significant role that social relationships and networks play in knowledge creation, transfer, absorption and utilization (Anderson and Parker, 2013), and scholars refer to such networks as knowledge networks (Chen *et al.*, 2022). A knowledge network is a collection of individuals and groups that transcend organizational, spatial and disciplinary boundaries, enabling the sharing of knowledge systems. It creates value by aggregating and implementing knowledge through the capitalization of intellectual capital (Abualqumboz *et al.*, 2021). Some scholars predict that research on intellectual capital tends to extend the boundaries of institutions as stakeholders in a larger ecosystem to help address social problems (Secundo *et al.*, 2018a, b). Hence, it holds great practical significance to investigate the intellectual capital in corporate universities based on the knowledge network in this paper, which can not only enhance the capability of enterprise management talents but also promote the sustainable development of enterprises.

It is evident that the synergy and intercommunication among the different elements of intellectual capital form the foundation and bridge that affects value performance. However, most previous studies on the various elements of intellectual capital have analyzed them from a relatively independent and simplistic linear perspective (Festa *et al.*, 2021). Corporate universities possess organizational characteristics of openness and cross-border penetration, enabling them to balance and coordinate internal and external resources for the establishment of a knowledge network that serves enterprise knowledge management. Intellectual capital plays a vital role as a source of competitive advantage within the knowledge network of corporate universities. The internal elements of intellectual capital are intricately intertwined with external capital and also possess multi-dimensional network attributes (Chen *et al.*, 2022). Therefore, it is imperative to examine the relationship formed through mutual cooperation and connection among the diverse elements of intellectual capital in corporate universities from a network perspective.

Previous studies have demonstrated the dynamic nature of knowledge networks and intellectual capital, suggesting that they should be considered as evolving rather than static entities (Dumay and Garanina, 2013; Abualqumboz *et al.*, 2021). Secundo *et al.* (2018b) emphasized the significance of considering different stages of intellectual capital development and its application across various contexts. In practice, the activities carried

out by corporate universities are becoming increasingly diverse. As the knowledge interface of enterprises, they effectively overcome interface obstacles and serve as centers for governing knowledge networks within enterprises (Patrucco *et al.*, 2017; Chen and Xu, 2022). As enterprises evolve, the requirements for the intellectual capital of corporate universities are bound to change dynamically. However, most existing studies have only described individual knowledge activities and supporting capabilities of corporate universities from a static perspective (Scarso *et al.*, 2017; Slettli and Grønhaug, 2017), without adequately considering the interaction between diversified intellectual capital and the knowledge networks of corporate universities, as well as the process of enrichment and expansion of these activities.

From the perspective of intellectual capital and knowledge networks in corporate universities, this paper uses grounded analysis and data processing technology for analysis. Taking three corporate universities in China as examples, this study further clarifies the constituent dimensions, mechanisms and evolutionary paths of intellectual capital in corporate universities based on the knowledge network.

From this perspective, the following major research questions are raised:

- RQ1. What is the connotation of intellectual capital and intellectual capital network within corporate universities?
- RQ2. What role does the intellectual capital network constructed by corporate universities play?
- RQ3. How does knowledge network-based intellectual capital evolve in corporate universities?

In summary, this study fills the above research gaps and makes certain contributions. First, different from previous studies which only studied simple linear relationships, this paper analyzes the relationship among the various elements of intellectual capital from a new perspective of knowledge networks. Second, existing scholars have rarely focused on the mechanism of intellectual capital in the knowledge management of corporate universities. Therefore, it is necessary to study the evolutionary process and synergy mechanism of intellectual capital network and knowledge management in corporate universities. Third, this paper addresses the lack of systematic exploration of the synergies between internal and external elements of intellectual capital in prior studies.

In the following chapters, the paper will first introduce the theoretical background and then describe the methodology used and the results obtained. Finally, it will present the discussion, conclusions, implications and limitations of the study.

2. Literature review

2.1 Knowledge network-based intellectual capital

Abualqumboz *et al.* (2021) proposed the concept of network-based intellectual capital, which expands from the individual level to encompass the entire social system by describing the process of value creation. It serves as a means to connect individuals, information and expertise in a complex and dynamic social network. Intellectual capital is an organizational phenomenon that represents the enterprise's ability to create wealth by effectively integrating various knowledge elements within a specific enterprise (Kianto *et al.*, 2017; Shet *et al.*, 2021; Marinelli *et al.*, 2023). Furthermore, it has been observed that the concept of intellectual capital stems from research on the interplay between intangible assets and performance, revealing a substantial positive correlation between intellectual capital and performance (Fiano *et al.*, 2020). Intellectual capital can be divided into three categories: human capital, structural capital and relational capital. The effective utilization of these

elements serves as the foundation of sustainable competition for enterprises (Bontis, 1998; Kianto *et al.*, 2017). Throughout the development of intellectual capital, it has primarily been examined as a measurable resource for value creation. Consequently, many scholars have focused on measuring and reporting intellectual capital, believing that both aspects serve as key drivers for enhancing business management (Lerro *et al.*, 2014; Liebowitz and Suen, 2000; Kianto *et al.*, 2014; Fiano *et al.*, 2020). Despite the extensive research on intellectual capital, few scholars have explored it from the perspective of a knowledge network.

Intellectual capital is characterized by its discrete, distributed and networked within the enterprise (Kianto *et al.*, 2017). Therefore, the interconnectedness of various components of intellectual capital can also be studied from a network perspective. Bontis (1998) argues that intellectual capital demonstrates network attributes, whereby diverse knowledge elements are interconnected and interacting within the organization's knowledge network. This intricate knowledge flow and transfer contribute to the development of a knowledge network-based intellectual capital.

However, there are challenges in governing the knowledge network and intellectual capital of enterprises. First, enterprises face difficulties in converting their human capital into organizational capital (Abualqumboz *et al.*, 2021). Knowledge workers possess an increasing number of intangible intellectual resources and skills, which often pose challenges in terms of expression and compilation, thereby impeding knowledge sharing and dissemination within organizations (Scarso *et al.*, 2017). Second, as enterprises progress and evolve, knowledge becomes progressively embedded within their operational processes and intricately intertwined with the innovation process, presenting challenges to the integration of intellectual capital (Petr and Elena, 2018). Moreover, enterprises encounter obstacles in searching for and acquiring relational capital that aligns with their specific needs, thus impeding the effective connection between externally scattered knowledge and internal knowledge (Chen and Xu, 2022). To address these issues, organizations need to establish a knowledge network that facilitates the management and connection of dispersed intellectual capital both inside and outside the enterprise.

Scholars argue that intellectual capital should be viewed as a dynamic entity in constant evolution (Dumay and Garanina, 2013). Research on intellectual capital has progressed through four stages, with the focus shifting accordingly at each stage (Secundo *et al.*, 2018b). In the current era of the knowledge economy and the expanding influence of social networks, intellectual capital has entered the fourth stage of research, which incorporates the perspective of ecosystems and considers the social dimension of stakeholders (Secundo *et al.*, 2018b). Nowadays, with the continuous development of knowledge networks, knowledge and information have become the competitive advantage of intellectual capital and the main drivers of value creation (Gabriele *et al.*, 2018; Fiano *et al.*, 2020; Festa *et al.*, 2021 Marinelli *et al.*, 2023).

2.2 Corporate university

Several scholars have recognized the potential of corporate universities in establishing effective knowledge management systems for enterprises, enabling the transformation and integration of diverse knowledge types into core resources. Consequently, corporate universities serve as organizations that provide comprehensive knowledge services to enterprises (Scarso *et al.*, 2017; Chen *et al.*, 2019). Previous studies have demonstrated the positive impact of corporate universities on knowledge management of enterprises. However, the specific knowledge activities undertaken by corporate universities and their relationship with internal and external knowledge resources remain unclear, leading to debates regarding the role of corporate universities in knowledge management and restricting efforts to enhance their capabilities.

Some scholars argue that corporate universities should primarily focus on training new employees in enterprise knowledge convergence and providing continuing education for on-

the-job personnel, essentially fulfilling the function of knowledge transfer within enterprises (Petr and Elena, 2018). Corporate universities act as platforms and mediums for knowledge networks and technological innovation of enterprise (Gabriele *et al.*, 2018). Nevertheless, it is crucial to recognize that knowledge exchange and sharing are processes within knowledge networks, whereas knowledge creation and aggregation represent the ultimate goals (Chen *et al.*, 2019). Therefore, corporate universities should not only undertake the education and training functions of knowledge and skills but also assume the role of knowledge interface to manage knowledge networks.

Furthermore, some scholars contend that corporate universities should actively engage in activities related to the knowledge development, collaboration and maturation of enterprises (Patrucco *et al.*, 2017; Chen *et al.*, 2022). A knowledge network represents an interactive system of nodes engaged in symbiotic knowledge relationships (Anderson and Parker, 2013), and corporate universities can serve as knowledge interfaces to facilitate knowledge dissemination and innovation activities (Slettli and Grønhaug, 2017). However, this perspective merely categorizes the functions of corporate universities based on knowledge management theory, focusing on specific knowledge activities such as acquisition, integration and application, while failing to fully explore the interconnectedness and holistic nature of knowledge activities of corporate universities.

Although some scholars have recognized that the knowledge management process of corporate universities can effectively facilitate the transfer and sharing of internal knowledge, thereby enhancing enterprise innovation performance through the integration and utilization of external knowledge (Chen and Xu, 2022), they often neglect the distinct characteristics of corporate universities in managing enterprise knowledge networks. Additionally, there is a lack of research on the knowledge interface function of corporate universities in combining knowledge processes for enterprises from the perspective of knowledge networks.

3. Methodology

The primary objective of this study is to investigate the knowledge network-based intellectual capital and its evolution mechanisms in corporate universities. This exploration is conducted using an exploratory case study approach, as the existing research theories in this field are not yet well-defined. Exploratory case studies are particularly suitable for conducting in-depth analyses of the theoretical underpinnings of problems and phenomena (Yin, 2003). The significance of case studies lies in their capacity to provide a foundation for the development of new theories. Moreover, exploratory case studies often transcend established theoretical frameworks by employing new perspectives, assumptions and methodologies to analyze social and economic phenomena (Yin, 2003). Therefore, given the aforementioned research questions regarding the “what” of RQ1 and RQ2, as well as the “how” of RQ3, adopting the case study method is appropriate.

3.1 Case selection

This paper has selected three corporate universities in China, namely Hisense College, Midea College and TCL Leadership Development College, as the samples for exploratory case studies. The reasons for selecting these three corporate universities are as follows: First, these institutions have relatively rich public information available, providing researchers with direct access to relevant data. Second, the chosen samples are widely recognized as highly successful benchmark corporate universities in China. They have clear stages of development, abundant retrospective data and undergone extensive and comprehensive knowledge activities, making them suitable for evolutionary analysis. Third, as knowledge

management in this study primarily emphasizes technology and knowledge processing, corporate universities established by enterprises in the technical field were selected. These enterprises are involved in complex technological innovations, possess rich industrial chains and have a strong demand for knowledge creation. Consequently, the knowledge activities and services provided by these corporate universities are more diverse.

3.2 Data collection

To ensure the reliability and validity of the case studies, the data collection and analysis followed the recommendations of [Yin \(2003\)](#). Multiple data collection methods were utilized, including semi-structured interviews, field research and the compilation of second-hand data. The data were then analyzed using triangulation techniques, where data from different sources were cross-verified to ensure the integrity and authenticity of the information obtained. The specific data sources are shown in [Table 1](#).

The investigation, interviews and analysis processes were carried out collaboratively by a research team comprising three main researchers. Following the completion of interviews, the research team members collectively discussed and exchanged the collected data and research findings. See [Table 2](#) for interviews. Through data collection and semi-structured interviews, a corpus of approximately 90,000 words of recorded manuscripts was compiled from 19 senior managers affiliated with the three corporate universities. Moreover, around 80,000 words of official materials and industry articles, as well as 40,000 words of manuscripts from four industry experts were gathered. The mutual verification of multiple data sources not only provides robust support for this study but also enhances the accuracy of constructing the theoretical model.

3.3 Milestone events of cases

Based on interview data and secondary sources, the milestone events in the development process of the three case subjects have been carefully examined and refined, as shown in [Table 3](#), [Table 4](#) and [Table 5](#).

3.4 Data analysis

One of the primary objectives of this study is to develop conceptual models, so it is necessary to refine and summarize case data and extract aggregated constructs. The analysis of the

	Data sources	Data catalogue
First-hand data	Semi-structured interviews	Semi-structured interviews were conducted with a total of 29 senior executives from corporate universities, who were requested to provide and validate pertinent information
	Consult with industry experts	Consultations were conducted with four industry experts, including the Deputy Director of the Research Department at Dalian Business Executives Academy, a Professor from the Overseas Education College at Shanghai Jiaotong University, the Manager of the Business Department at GLO Consulting Company and a Consultant from the Research and Training Center of a corporate university
Second-hand data	Field investigation	Corporate university training center, enterprise exhibition hall, etc.
	Internal information External data	Work reports and enterprise chronicle, etc. Official websites, media reports, interview records of their presidents, relevant academic literature and books

Source(s): Interview records

Table 1.
Data sources

Table 2.
Job positions of the interviewees, content and hours of interview

Position	Times of interviews	Total interview time (hours)	Interview questions
Executive president (A)	3	5.5	• How can the recruitment and development of teachers in corporate universities be effectively carried out?
Vice president (B)	2	4.5	
General manager (C)	2	5	
Director of the department of training and planning (A)	1	2.5	• Are corporate universities closely connected with various business departments of enterprises? • How do corporate universities introduce and evaluate external knowledge of enterprises? • How do corporate universities transfer the accumulated knowledge, protocols and experience of enterprises to employees? • How do corporate universities facilitate the integration of subject education and enterprise knowledge for employees?
Director of the department of internal training and management (B)	1	2	
Director of the department of internal training (C)	1	2	
Director of department of external exchange (A)	2	4.5	• How do corporate universities transfer the accumulated knowledge, protocols and experience of enterprises to employees? • How do corporate universities facilitate the integration of subject education and enterprise knowledge for employees?
Director of external training (B)	1	2.5	
Director of the department of external training (C)	1	2	
Director of department of learning and development (A)	1	2	• How do corporate universities facilitate the integration of subject education and enterprise knowledge for employees?
Director of research and development office (B)	2	4	
Director of subjects (C)	2	4.5	
Director of the department of production (A)	1	2	• How do corporate universities facilitate the integration of subject education and enterprise knowledge for employees?
Manager of project management department (B)	2	4	
Director of operations (C)	1	1.5	
Marketing manager (A)	1	2	• How do corporate universities facilitate the integration of subject education and enterprise knowledge for employees?
Marketing manager (B)	1	2	
Merchandising manager (C)	1	2.5	
Deputy dean of the college (A)	3	5	

Note(s): “A” stands for Hisense College, “B” stands for Midea College, “C” stands for TCL Leadership Development College

Source(s): Interview records

Table 3.
Milestone events of Hisense College

Time	Milestone events
1998	It was established. It is one of the earliest local corporate universities in China
2006	With the acquisition of Kelon Company by Hisense Group, both the staff and business volume of the group experienced a significant increase. As a result, the training scope was expanded further and over 5,000 staff members were trained annually
2015	It established a management research center, which aims to achieve cross-border research by building a platform that connects with internal business departments of the enterprise
2022	Hisense Group has established several research and development centers, state-level key studios, research institutes and postdoctoral mobile stations. It serves as the primary coordinating unit for these collaborative institutions, signifying the increasing emphasis placed by Hisense College on fundamental theoretical research and engineering scientific research activities in the industrial domain

Source(s): Interview records

Table 4.
Milestone events of
midea college

Time	Milestone events
2006	It was founded by Midea Group in 2006. Its main training programs include the New Employee Training Camp, Reserve Manager Training Camp, New Manager Training Camp, Leadership Development Training and Overseas Training, so as to cultivate comprehensive professional talents aligned with Midea's internationalization strategy
2014	It is dedicated to serving as a strategic partner for Midea Group's business development strategy, enterprise management research center and employee innovation activity platform
2019	As an intermediary, it strengthens the technical cooperation and development among enterprises, universities and research institutes, participates in fundamental collaborative projects between Midea research institute and technology center, placing significant emphasis on cultivating employees' innovative ability
2022	By providing technical services to enterprise to promote technological innovation and building extensive network connections for existing resources such as market, supplier relationships, technology and brands, it has transformed into a collaborative platform for technological innovation and R&D within Midea Group

Source(s): Interview records

Table 5.
Milestone events of
TCL leadership
development college

Time	Milestone events
2000	Established in 2000, it is the strategic planner for talent development and the integrator of training resources, focusing on providing management knowledge and skills training for middle and senior managers in the Group
2006	It maintains a business guidance and support partnership with the training departments across all branches of the group, collaborating on various projects including new employee training
2012	It has integrated the internal practice cases of the Group into the college's curriculum, playing a dual role in accumulating and disseminating knowledge
2015	It has implemented internationally advanced training courses and fostered deeper collaborations with renowned business schools and training consulting institutions, such as China Europe Business School, Guanghua Management School of Peking University and the University of Texas
2022	It serves as not only a comprehensive training college for the entire group, but also a dynamic platform that facilitates interaction, discussions and communication among employees

Source(s): Interview records

case data structure follows the three procedures proposed by [Strauss and Corbin \(1998\)](#) for research utilizing the grounded theory method, which includes first-order concepts, second-order themes and aggregate construction.

During the open coding stage, the formation of first-order concepts involved coding, summarizing and labeling relevant phrases or paragraphs in the original data line by line and sentence by sentence. From the original data of Hisense College, a total of 325 coding words and phrases were extracted. Similarly, 297 codes were extracted from the original data of Midea College, and 314 codes were derived from the interview records and documents of TCL Leadership Development College. Subsequently, we identified 92 concept labels shared by all three corporate universities.

In the axial coding stage, second-order themes were derived from a collection of related first-order concepts, thereby reflecting their interrelationships and underlying patterns. Through repeated comparisons with pertinent literature, similarities and differences between the new findings and existing theories were identified. The 19 initial categories were eventually reclassified into 6 subcategories. The coding results are shown in [Table 6](#).

The researchers conducted further refinement and summarization of the aggregated concepts based on the second-order themes, which contributed to the theoretical construction of

Codes	19 categories	6 sub-categories
31: The senior management of the Group, including the Party Secretary and General Manager, consistently delivers lectures to the participants	The level of teachers in corporate universities	Internal teachers of the corporate universities
32: Interaction and complementary relationship in terms of teaching content between the middle and senior faculty members	The relevance of the content taught by teachers within the corporate universities	
11: Excellent talents are selected within the group to teach the employees	Cultivation of internal teachers	
38: The trainers within the group possess vast expertise and extensive experience in their respective technical fields		
15: Externally recruited faculty members demonstrate a strong ability to communicate the strategic intentions of senior management when teaching	Familiarity of external teachers of corporate universities with enterprises	External teachers employed by corporate universities
54: External R&D experts and technical experts are hired as part-time teachers	The level of external teachers of corporate universities	
42: The proportion of external lecturers at the college is determined according to the training orientation and level of the trainees	Proportion of external teachers of corporate universities	
45: Conduct an examination to identify the specific knowledge areas that the enterprise needs to master and assess the existing knowledge gaps within the various business departments	Explore the knowledge gap of enterprises	Sorting out the knowledge process within the enterprises
81: Identify knowledge gaps		
47: Gain a comprehensive understanding of the processes and fundamental tasks carried out by each operating department	Cross-interface knowledge convergence	
07: Attend regular meetings with various departments to engage in business discussions and identify learning needs	Acquiring matching knowledge	
02: Exploring cutting-edge knowledge in the industry		Sorting out the external knowledge process of enterprises
03: Conducting a series of feasibility studies on the requirements of overseas markets and potential investments abroad	Evaluate the external knowledge of enterprises	
28: Application of the advanced management ideas summarized and refined within the Group	Internalization of external knowledge of enterprises	
68: Participation in industry associations, corporate university consortia and other forums to strengthen technical exchanges and establish mutual visits	Connection of enterprise external knowledge	
05: Exchange of cases at the college and application of the experience gained in daily work	Tacit knowledge is transformed into explicit knowledge	Organizational learning mechanism
83: Develop and explore the mobile terminal learning platform and information construction of the college	Information system construction and application	

Table 6.
Examples of open
coding and axial
coding

(continued)

Table 6.

Codes	19 categories	6 sub-categories
22: Built an online learning platform to provide comprehensive knowledge services for employees, customers and partners	Online learning resources	
90: Assemble a project team for a technical study, dedicated to conducting collaborative research initiatives	Internal communication mechanism of the enterprise	
84: Most courses allocate a significant portion of time for free discussions, fostering an optimal atmosphere for engaging and dynamic participant interactions	Create a psychological environment	
19: Teamwork projects designed to increase staff communication skills	Harmonize the relationship among students	
75: Organize knowledge competitions and various activities to foster the development of practical skills among staff members	Teaching and auxiliary activities	
Note(s): Only some codes are provided in the table as examples		
Source(s): Interview records		

the case data. In the selective coding stage, repeated comparisons were conducted to elucidate the internal relationship between the aforementioned concepts and facilitate deeper exploration.

By selective coding and further refining the six aggregated concepts derived from the previous analysis, noticeable distinctions among these concepts were identified through comparison. Ultimately, three main categories were summarized based on their relationships: teacher network capital, knowledge process capital and knowledge ba (see [Table 7](#)).

Core category	Categories	Key word coding of the case data
Teacher network capital	Internal teachers of the corporate universities	The level of teachers in corporate universities; The relevance of the content taught by teachers within the corporate universities; Cultivation of internal teachers
	External teachers employed by corporate universities	Familiarity of external teachers of corporate universities with enterprises; The level of external teachers of corporate universities; Proportion of external teachers of corporate universities
Knowledge process capital	Sorting out the knowledge process within the enterprises	Explore the knowledge gap within enterprises; Cross-interface knowledge convergence; Acquiring matching knowledge
	Sorting out the external knowledge process of enterprises	Evaluate the external knowledge of enterprise; Internalization of external knowledge of enterprises; Connection of enterprise external knowledge
Knowledge ba	Organizational learning mechanism	Tacit knowledge is transformed into explicit knowledge; Information system construction and application; Online learning resources
	Knowledge sharing platform	Internal communication mechanism of the enterprise; Create a psychological environment; Harmonize the relationship among students; Teaching and auxiliary activities
Source(s): Constructed by the authors		

Table 7.
Coding results of
intellectual capital of
corporate universities

4. Findings

RQ1: What is the connotation of intellectual capital and intellectual capital network of corporate universities?

4.1 Intellectual capital and intellectual capital network of corporate universities

4.1.1 Intellectual capital of corporate universities. According to the case study, it is evident that the intellectual capital of corporate universities depends on the knowledge, skills and experience accumulated by teachers, and students within these institutions. This is manifested in various aspects such as the knowledge sorting process, information system, organizational structure, coordination mechanism and internal and external association mechanism of corporate universities. These elements represent the connection mode among process capital, knowledge capital, human capital and relationship capital, providing support for knowledge transfer, knowledge creation and the enhancement of corporate university capabilities. Through the examination of corporate universities, it is observed that the dimensions of intellectual capital are embodied in teacher network capital, knowledge process capital and knowledge ba. The different intellectual capitals of corporate universities can be regarded as a combination of intellectual capitals. It is important to note that the intellectual capital of corporate universities does not simply involve an additive relationship. Instead, these components collaborate and synergize to promote the enhancement of knowledge management abilities in corporate universities. Based on the data coding, this paper introduces a combination model of the intellectual capital of corporate universities, which is referred to as the “steering wheel” model of intellectual capital combination of corporate universities (see Figure 1).

(1) Teacher network capital

The teacher network capital represents a fundamental and significant resource of corporate university. It encompasses both full-time and part-time teachers employed within the enterprise, as well as external part-time teachers, which can be further classified as internal teacher network capital and external teacher network capital. Table 8 for the description of

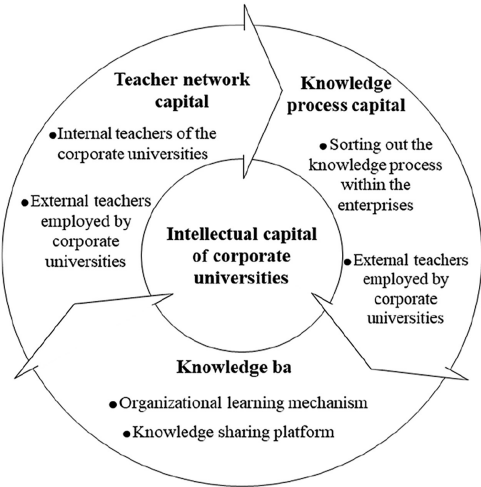


Figure 1.
“Steering wheel” model
of the combination of
intellectual capital of
corporate universities

Source(s): Authors work

		Knowledge network-based intellectual capital
Cases	Interview data	
Hisense College	It has a faculty of over 200 full-time training teachers, with external teachers constituting 20% of the total. Additionally, nearly 500 middle and senior managers, R&D experts and technical professionals from internal enterprise certifications serve as part-time instructors. It is dedicated to enhancing the professional development of its teaching staff, ensuring that all technical trainers possess extensive experience in their respective fields of expertise	1615
Midea College	The teaching staff at Midea College primarily consists of both full-time and part-time teachers who are employed within the enterprise, with limited external teachers. The faculty of Midea College maintains a moderate scale, comprising over 100 trainers certified overseas and more than 300 full-time teachers. Additionally, there is a substantial number of part-time teachers from internal management and business departments of various enterprises, reaching several thousand in total	
TCL leadership development College	It has a distinguished faculty consisting of over 100 certified teachers from various departments of the enterprise. In order to align with the training objectives for employees at all levels, the college strives to maintain an equitable ratio of internal and external lecturers. The teachers selected by College are part of a dynamic teacher rotation system, rather than a lifelong appointment, thereby ensuring a well-regulated flow of educators	
Source(s): Interview records		Table 8. Examples of interview data of teacher network capital

the interview information related to teacher network capital in the three corporate universities.

(2) Knowledge process capital

Based on the case study, it was found that the knowledge process capital of corporate universities can be divided into internal knowledge process capital and external knowledge process capital.

Internal knowledge process capital involves utilizing specific methods and experiences of knowledge management to handle internal business processes within the enterprise. This includes establishing appropriate methods and mechanisms for storing and accumulating knowledge generated across various stages and departments within the enterprise. On the other hand, external knowledge process capital refers to the acquisition of information, knowledge and skills through external networks. Interview data from the three corporate universities regarding knowledge process capital are presented in [Table 9](#).

(3) Knowledge ba

Corporate universities serve as centralized hubs for gathering knowledge resources, human resources and technical resources. They create an environment and atmosphere that are conducive to knowledge sharing and creation, which contributes to the concept of knowledge ba. The knowledge-sharing platform established by corporate universities not only effectively facilitates knowledge exchange and interaction among employees within the enterprise but also accumulates and enriches diverse knowledge among organizations. Additionally, they enable the mutual transformation of tacit knowledge and explicit knowledge within and outside the enterprise. See [Table 10](#) for the interview data.

4.1.2 Knowledge network of intellectual capital of corporate universities. Knowledge management in corporate universities is recognized as an integrative process that combines intellectual capital and facilitates the flow of knowledge. A comprehensive utilization of various forms of intellectual capital leads to effective knowledge management, which is the result of the overall operation of intellectual capital in corporate universities. Within the

Table 9.
Examples of interview
data of knowledge
process capital

Cases	Interview data
Hisense College	Hisense Group diligently implements a comprehensive strategy of internationalization and intelligence, while Hisense College places great emphasis on addressing internal challenges faced by the enterprise. The college systematically synthesizes and distills the successful practices of Hisense, incorporating cutting-edge theories and experiences from diverse nations to cultivate distinctive management principles, so as to elevate the overall management capabilities of the entire group Hisense Group endeavors to attain a more forward-thinking comprehension of knowledge and information to facilitate the expansion of its overseas business. Concurrently, it can efficiently harness its organizational advantages in cross-border operations and knowledge exchange, while conducting an extensive array of feasibility studies on overseas market demands and investment opportunities
Midea College	The Institute of Management established by Midea College provides consulting services for the Group's continuous operation, management transformations and decision-making processes. Moreover, it conducts systematic analysis and enhancement of the internal management tools, methodologies and best practices employed by the organization It has established a strong connection between enterprise, famous universities and scientific research institutions, facilitating the introduction of novel technologies and ideas to promote the effective transformation and application of scientific and technological achievements
TCL leadership development College	It actively engages in cross-departmental meetings to facilitate comprehensive business discussions, gaining insights into the operational procedures and key responsibilities of various business units. Furthermore, it designs tailored learning initiatives for business departments, focusing on enhancing their expertise in specialized projects It organized industry information sessions and international exhibitions, effectively capturing an extensive array of cutting-edge knowledge and internationally recognized technical certification standards in the field of intelligent manufacturing. This valuable intelligence was subsequently integrated into the internal development and design processes of enterprises, ensuring their alignment with international best practices
Source(s): Interview records	

knowledge network of corporate universities, intellectual capital is externalized as knowledge node. These nodes establish innovative and stable relationships with one another through interactions and collaborative innovation, enabling the divergent flow and transfer of the knowledge chain. Intellectual capital itself possesses multidimensional network properties and serves as a crucial source for enhancing the knowledge management capabilities of corporate universities.

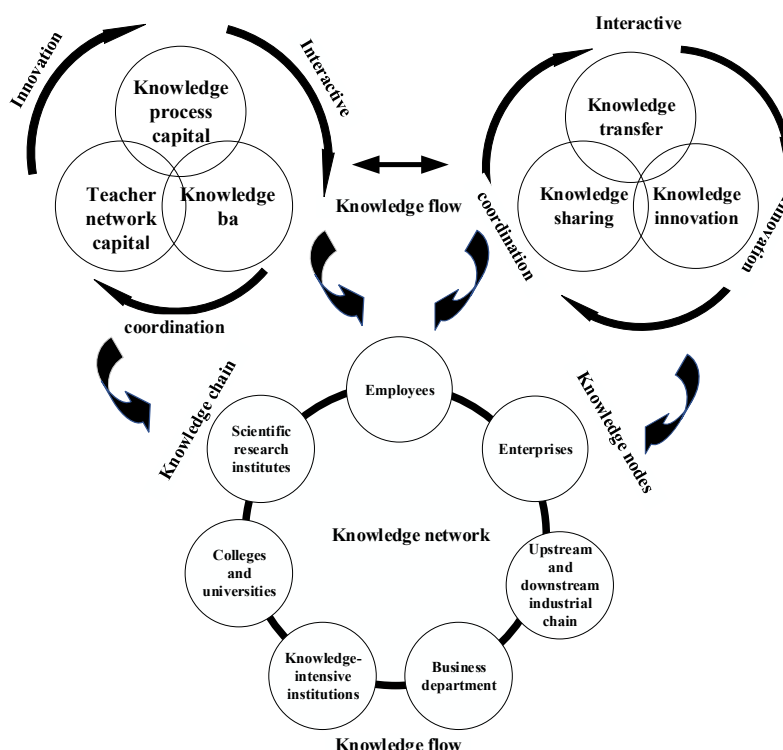
However, it should be noted that the advancement of knowledge management in corporate universities is not solely dependent on the three independent components of intellectual capital but rather realized through their dynamic interaction. The knowledge interaction and knowledge network among the teacher network capital, knowledge process capital and knowledge ba of corporate universities constitute the knowledge network of intellectual capital. Consequently, the knowledge management activities undertaken by corporate universities are based on the knowledge network of intellectual capital. [Figure 2](#) depicts the model of the intellectual capital network of corporate universities.

There are two primary forms of knowledge flow in the intellectual capital network of corporate universities. First, there is the knowledge flow between the intellectual capital of

Table 10.
Examples of interview
data of knowledge ba

Cases	Interview data
Hisense College	Hisense College periodically convenes core partner consortiums to engage in deliberations on cross-industry collaboration pertaining to intelligent technology with operators and prominent industry leaders
Midea College	Employees are encouraged to actively contribute and share their business cases on the dedicated platform provided by the college. This allows for challenging problems encountered in their work to be brought forth during seminars, fostering classroom discussions. Furthermore, these cases can be recommended to company managers through the platform, facilitating widespread circulation. Notably, exceptional cases compiled by trainees are acknowledged and rewarded, subsequently being utilized in future training programs. Additionally, employees' test responses, scores and self-evaluations are posted on the college's website after each class, providing a convenient means for independent learning. Based on performance in exams or during class activities, outstanding employees can be identified and recommended for promotion
TCL leadership development College	The college has built a knowledge community and a mobile learning platform for employees. It also integrates various interactive forms such as forums, synchronous classes and chat rooms and takes this platform as an ideal medium to promote knowledge sharing among the employees

Source(s): Interview records



Source(s): Constructed by the authors
Authors work

Figure 2.
Model of intellectual
capital network of
corporate university

the corporate university and the internal nodes within the enterprise. Second, there is the knowledge interaction that occurs between the corporate university and the external nodes of the enterprise. Intellectual capital in corporate universities undergoes knowledge flow and transformation, forming an internal network of intellectual capital. Leveraging its advantages as a knowledge intermediary, the corporate university not only facilitates interactive transformation of intellectual capital within the enterprise but also establishes connections with various external nodes, thereby constructing an external intellectual capital network. The integration of internal and external intellectual capital networks in corporate universities empowers and promotes knowledge innovation among technicians, teams, business departments, as well as the upstream and downstream industry chains of enterprises. The connections and coordination of the knowledge network, facilitated by teacher network capital, knowledge process capital and knowledge ba in corporate universities, play a crucial role in aggregating innovative resources and ensuring the efficient operation of enterprise knowledge activity systems. Examples of interview data for the intellectual capital networks of corporate universities are shown in [Table 11](#).

RQ2: What role does the intellectual capital network constructed by corporate universities play?

Cases	Interview data
Hisense College	Hisense College actively contributed to the development of multiple industry standards, introducing cutting-edge information and international technical certification standards to enterprises. These standards were effectively utilized in product development and design processes. Over time, the college has evolved into a knowledge-intensive institution, playing a pivotal role in guiding and overseeing the training initiatives within Hisense Group. It also collaborates with various research institutions affiliated with Hisense Group, integrating both internal and external resources. As a result, it has emerged as a comprehensive platform for training, research and innovation incubation within Hisense Group
Midea College	The President of Midea College holds the position of Chairman within Midea Group, while the Vice President serves as a member of the board of directors and assumes the role of Chief Learning Officer. This direct involvement in enterprise-wide strategic formulation enables effective coordination among various departments. Consequently, the planning developed by the corporate university is strategically aligned. Midea College transcends traditional training organizations by establishing extensive connections with diverse knowledge departments within Midea Group
TCL leadership development College	Through combing the internal and external knowledge processes of the enterprise, it can explore the knowledge weaknesses of various departments of the enterprise, track the development trends of the industry, seek matching knowledge information, integrate the internal knowledge of the enterprise and the knowledge of upstream and downstream industrial chains and promote the development of knowledge management activities of Group By systematically combing the internal and external knowledge processes of the enterprise, it is able to identify knowledge gaps within different departments. It actively tracks industry development trends, seeking relevant knowledge information to address these gaps. Furthermore, it integrates internal knowledge resources in the enterprise with knowledge from upstream and downstream industrial chains. Through these efforts, it plays a pivotal role in promoting the advancement of knowledge management activities within the Group

Table 11.
Examples of interview data for intellectual capital networks in corporate universities

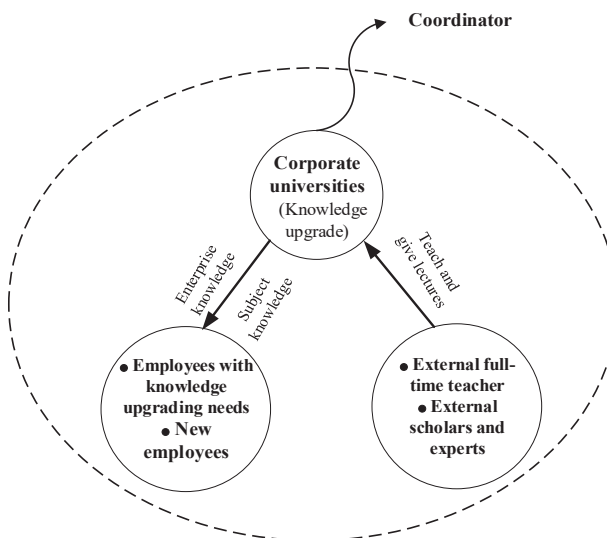
Source(s): Interview records

4.2 Role of the network of intellectual capital in corporate universities

The intellectual capital within the knowledge network of corporate universities serves as an intermediary, facilitating the acquisition and integration of diverse knowledge and information by enterprises. According to the distinct characteristics of intellectual capital in corporate universities, the roles of intellectual capital within the knowledge network can be summarized as follows: the intellectual capital network with the role of coordinator, the intellectual capital network as the knowledge gatekeeper and the intellectual capital network as the innovation bridge. The roles and characteristics of various intellectual capital networks are analyzed below.

4.2.1 Coordinator. The intellectual capital network of corporate universities assumes the role of coordinator, utilizing the knowledge network of intellectual capital to establish an educational platform that facilitates training and knowledge exchange within the enterprise. The training programs offered by corporate universities primarily focus on enhancing enterprise knowledge for new employees and providing post-education academic qualifications following knowledge upgrading. By studying and discussing the business environment, customer demand information, industry trends and enterprise strategic development direction, corporate universities facilitate the exchange and collaboration of knowledge and experience among engineers and employees in diverse positions and disciplines. In this process, the knowledge network of intellectual capital serves as the role of coordinator, as shown in Figure 3.

A typical example of an intellectual capital network that assumes the role of coordinator is the TCL Leadership Development College. During interviews with middle and senior managers from the college, they consistently emphasized the contribution of the corporate university in facilitating knowledge sharing and communication among employees. For instance, "... The college has developed a comprehensive curriculum system that focuses on values training, leadership development, professional skills enhancement, and career advancement. This system enables employees to acquire cutting-edge industry knowledge



Source(s): Constructed by the authors
Authors work

Figure 3.
Role of coordinator of
intellectual capital
network in corporate
universities

and upgraded subject expertise, thereby promoting the integration of subject education and enterprise knowledge”. “In order to promote knowledge exchange among employees within the enterprise, the college organizes small problem-solving seminars periodically. These sessions provide trainees with an opportunity to discuss and communicate in a relaxed atmosphere, fostering the establishment of informal relationships among employees ...”. “Through extensive training programs, lectures, the creation of a case sharing platform, business competitions and seminars, the college effectively harnesses the knowledge resources available across various business departments of the enterprise ...”.

4.2.2 Knowledge gatekeeper. The role of knowledge gatekeeper in a corporate university is to facilitate the assimilation of external knowledge into the enterprise. This involves acquiring external knowledge, transforming it into tacit knowledge within the enterprise, and internalizing the newly introduced knowledge. Corporate universities leverage cross-border advantages to search for information, evaluate external knowledge and introduce valuable knowledge resources. By doing so, they not only gather international technical certification standards or introduce high-quality courses from abroad but also monitor industrial technology trends and integrate relevant external knowledge into the enterprise’s knowledge base. Moreover, corporate universities play a crucial role in identifying external knowledge, filtering market information that has been gathered, and exploring the demands of customers beyond the enterprise. To facilitate technical exchanges between enterprises and external knowledge-intensive institutions within the same industry, they organize forums, lectures and conferences on industry development trends, serving as a platform for collaboration. Additionally, corporate universities offer tailored training services that cater to the upstream and downstream sectors of the industrial chains, thereby facilitating the integration of knowledge within these chains. The role of knowledge gatekeeper of intellectual capital network in corporate universities is illustrated in [Figure 4](#).

A prime example of an intellectual capital network that fulfills the role of a knowledge gatekeeper is Midea College. For example, “It collaborates with prestigious institutions such as

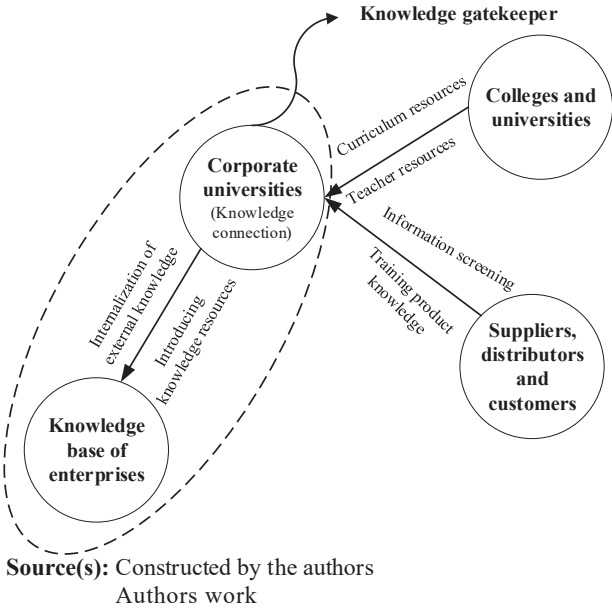
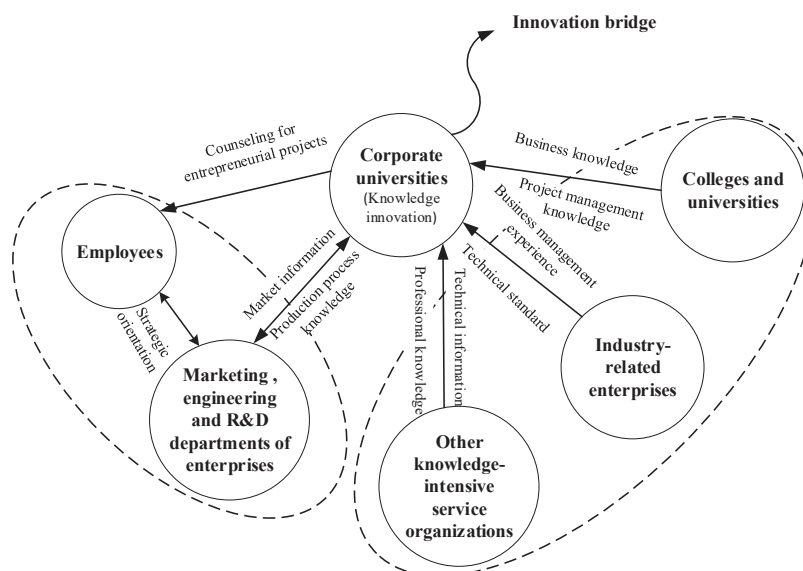


Figure 4.
Role of knowledge
gatekeeper of
intellectual capital
network in corporate
universities

Shandong University and the Massachusetts Institute of Technology, thereby enhancing the competence of its teachers and refining its curriculum system ...". "Midea College conducts various training programs focused on technical commercialization, providing technical evaluations and guidance on commercialization strategies for both upstream and downstream industrial chains as well as customers ...". "Through extensive information search, it collects market data, explores customer demands, and introduces advanced management methods and industry standards to enterprises. This endeavor has yielded significant achievements in terms of information screening and the identification of external knowledge ...". "By compiling international technical certification standards, it refines the supply chain management model and adopts the Six Sigma methodology ...". "Midea College annually sends hundreds of employees to renowned multinational companies such as Samsung and Hitachi to participate in management training and engage in technical collaboration ...".

4.2.3 Innovation bridge. The role of the intellectual capital knowledge network as an innovation bridge is evidenced by the promotion of interaction between internal and external knowledge networks within enterprises through corporate universities. Within the intellectual capital network of corporate universities, they gather the knowledge requirements of various business departments, identify knowledge gaps by analyzing business processes and further explore knowledge resources that align with these identified gaps. In the external intellectual capital network, they serve as interfaces for knowledge integration, leveraging resources such as universities and research institutes to engage in technical collaboration, thereby driving technological innovation within enterprises. The role of the intellectual capital network as an innovation bridge establishes both internal and external innovation platforms for enterprises. These platforms not only explore common knowledge, management theories and cutting-edge engineering science at the forefront of industries but also provide services to enterprises in terms of integrating intellectual resources, capital resources and technical resources for innovation incubation. The role of innovation bridge of intellectual capital network in corporate universities is depicted in Figure 5.



Source(s): Constructed by the authors
Authors work

Figure 5.
Role of innovation
bridge of intellectual
capital network in
corporate universities

A prime illustration of an intellectual capital network that takes on the role of an innovation bridge is Hisense College. During interviews with middle and senior managers at Hisense College, they repeatedly emphasized the institution's commitment to promoting the intelligent industry, fostering scientific research collaborations among universities, hosting high-level salon lectures and introducing emerging cutting-edge technologies. For example, "Hisense College aims to become the enterprise's hub for independent innovation, placing increasing emphasis on exploring industrial frontiers and conducting fundamental theoretical research ...". "... Hisense College relies on its post-doctoral workstation and industry-university-research alliance comprising six universities including Xi'an Jiaotong University to provide technical innovation consultations for the group and attract famous high-tech enterprises and projects". "Hisense College has developed into a comprehensive technology incubation base for the enterprise ...". "When employees have novel ideas, they are encouraged to submit project applications to Hisense College and form project teams freely. Through repeated discussions on ideas, feasibility assessment, market research and expert certification, Hisense College offers extensive support in terms of funding and technology for projects deemed worthy of implementation, thereby helping transform these ideas into reality ...".

RQ3: How does knowledge network-based intellectual capital evolve in corporate universities?

4.3 Evolution of the network of intellectual capital of corporate universities

4.3.1 Sequential inertia model of knowledge network-based intellectual capital of corporate universities. The intellectual capital and knowledge network contribute to the construction of knowledge network-based intellectual capital within corporate universities. The knowledge network-based intellectual capital is not static but rather represents a dynamic process characterized by interrelated and mutually reinforcing elements.

The knowledge network-based intellectual capital of corporate universities follows a two-way promotion path (see Figure 6). The establishment of high-level intellectual capital relies on the foundation provided by low-level intellectual capital. It is evident that corporate universities lacking teachers and adequate teaching resources would be ill-equipped to effectively integrate internal and external knowledge in enterprises. Conversely, the enhancement and advancement of low-level intellectual capital are also facilitated by high-level intellectual capital. In other words, the comprehensive integration of internal and external knowledge processes in corporate universities enables the aggregation of a broader

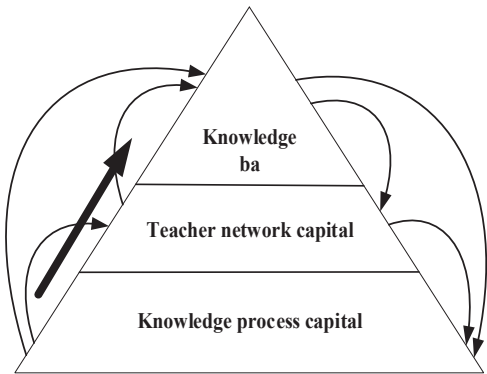


Figure 6.
Sequential model of
intellectual capital of
corporate universities

Source(s): Constructed by the authors
Authors work

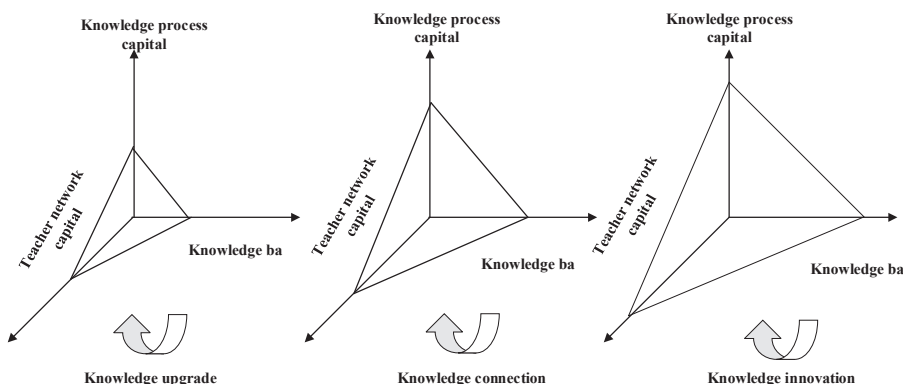
range of educational and knowledge resources, thereby enhancing the teacher network capital and knowledge ba of these institutions.

The case of Hisense College serves as a prime example of a mature corporate university, showcasing its comprehensive accumulation of intellectual capital. This accumulation extends beyond the realms of teacher network capital and knowledge ba to actively engage in the accumulation of knowledge process capital. In doing so, Hisense College expands both its internal and external knowledge groups, thereby constructing an intellectual capital network that assumes the crucial role of an innovation bridge. The three types of intellectual capital are coupled, and the network nodes within and outside the corporate university gradually expand, which strengthens the relationships between network nodes and gradually constructs an intellectual capital network with the role of an innovation bridge.

4.3.2 Collaborative evolution of knowledge network-based intellectual capital and knowledge management activities in corporate universities. From the perspective of resources and knowledge, intellectual capital plays a pivotal role in improving organizational capabilities. Its essence lies in fostering and establishing connections between knowledge activities, thereby increasing the efficiency of these activities within enterprises.

Through case analysis, a synergistic relationship between knowledge network-based intellectual capital and the knowledge activities of corporate universities has been identified. Initially, the knowledge activities of corporate universities are primarily focused on knowledge transfer, with relatively limited network connections. Corporate universities primarily establish relationships between students and teachers, prioritizing the development of teacher network capital to effectively support corresponding education and training activities. Knowledge connection and application incubation can be considered as behaviors of knowledge creation and utilization, which require the accumulation of higher-level cascade intellectual capital, namely, internal knowledge process capital and external knowledge process capital, in order to further promote and support the knowledge management capabilities of corporate universities. As corporate universities engage in knowledge connection activities, both knowledge ba and knowledge process capital gradually improve. In knowledge innovation activities, the three types of intellectual capital become intricately interconnected and continuously enriched (see Figure 7).

4.3.3 Ladder evolution of the intellectual capital and knowledge activities based on knowledge network in corporate universities. At different stages of corporate university development, the knowledge management activities undertaken by these institutions progress incrementally.



Source(s): Constructed by the authors
Authors work

Figure 7.
Collaborative evolution
model of intellectual
capital and knowledge
management activities
in corporate
universities

Consequently, the requirements for intellectual capital and intellectual capital networks in corporate universities vary accordingly. In the following section, this paper analyzes the evolutionary path of intellectual capital network and intellectual capital promotion matched with knowledge management activities at each stage. Figure 8 illustrates the ladder evolution of knowledge network-based intellectual capital in corporate universities.

(1) The stage of bridging education

During the stage of bridging education, knowledge upgrading serves as the primary activity of corporate universities. The network consists of a limited number of nodes, primarily facilitating face-to-face communication between teachers and students. In this stage, the network comprises only a limited number of nodes, primarily facilitating face-to-face communication between teachers and students. Corporate universities can fulfill the role of coordinator by aligning the intellectual capital network accordingly. Teacher network capital assumes a significant role in this stage. However, corporate universities predominantly rely on externally hired teachers due to the prevalence of part-time internal teachers within enterprises who often lack training experience. Consequently, internal teacher recruitment from within the enterprise is infrequent. At this stage, corporate universities mainly leverage their external knowledge process capital to introduce internationally recognized high-quality courses through the evaluation of external knowledge. Additionally, corporate universities have adopted mobile network education platforms and online information databases, utilizing diverse Internet learning resources and technologies as crucial development strategies.

An interviewee from TCL Leadership Development College said:

With the expansion of enterprise scale and the growing number of employees, it takes on the responsibility of introducing the values of new employees and improving the knowledge system of employees.

At this stage, the teacher network capital of TCL Leadership Development College is characterized by a limited number of internal teachers in the enterprise, prompting the college to primarily employ teachers from external sources.

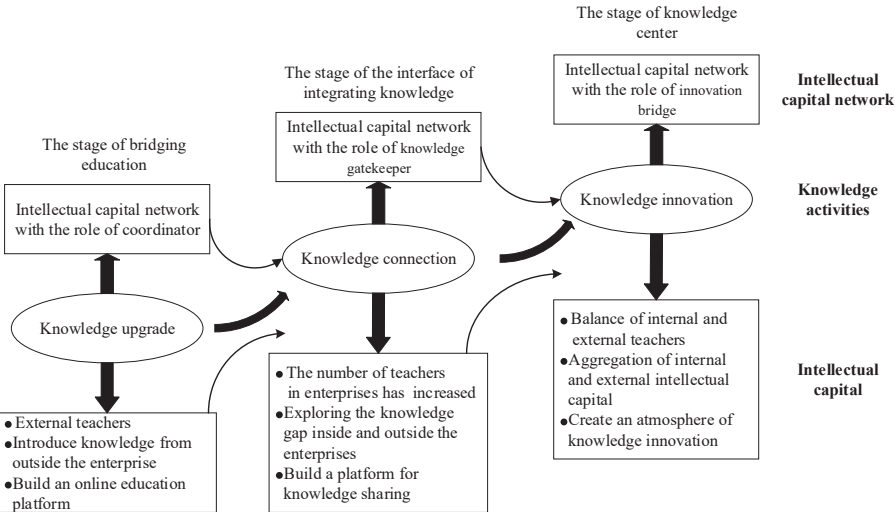


Figure 8.
Ladder evolution of
knowledge network-
based intellectual
capital in corporate
universities

Source(s): Constructed by the authors
Authors work

Another interviewee noted:

During the initial phases of TCL Group's development, the acquisition of knowledge primarily relied on external sources. As a result, TCL Leadership Development College often cooperates with foreign consulting institutions to develop case studies and assembles a team of teachers with expertise aligned with the business of the enterprise.

As a result, TCL Leadership Development College actively engages in collaborations with international consulting institutions to develop case studies. Additionally, it assembles a team of teachers who possess expertise that aligns with the business of the enterprise, thereby providing employees with comprehensive guidance on essential skills.

(2) The stage of the interface of integrating knowledge

In the stage of the interface of integrating knowledge, corporate universities establish the interfaces to connect internal and external knowledge of the enterprise. Corporate universities serve as platforms for knowledge sharing and communication among different business units and provide targeted training services to promote knowledge integration within the industrial chain. At this stage, corporate universities assume the role of knowledge gatekeepers, aligning their intellectual capital network accordingly. The network nodes have expanded beyond internal employees to encompass cross-departmental teams and business units. The external knowledge group comprises not only upstream and downstream partners in the industrial chain, but also related enterprises and research institutes. The focus of knowledge capital has shifted from constructing physical infrastructure to establishing a knowledge-sharing platform among employees. Within the intellectual capital network of corporate universities, the relationship between internal and external knowledge groups strengthens, leading to a gradual transition from expanding the external teacher network to expanding the internal teacher network. This transition is evident in the reduction of external teacher recruitment and the emphasis on improving the quality of teachers.

An interviewee from Midea College mentioned:

The composition of the teaching staff at Midea College has undergone a significant transformation, marked by a gradual decline in the number of external teachers and a substantial increase in the presence of internal teachers who possess expertise in business knowledge, industry knowledge and practical experience.

Board members and senior managers at Midea frequently give lectures to employees as part of their engagement with the learning process.

An interviewee provided us with information on this matter:

Midea College has established strong affiliations with research institutes and universities, leading to an expansion in the range of potential teachers. Collaborative efforts have resulted in the establishment of graduate workstations, laboratories, training centers, seminars and postdoctoral workstations.

The college has also embraced innovative teaching methods, departing from traditional classroom settings and adopting a more expansive training model that fosters an environment conducive to knowledge sharing.

This was corroborated by another interviewee:

It places great emphasis on accumulating green intelligent manufacturing technologies, encompassing equipment energy efficiency, system optimization and management strategies. By providing customers with energy-saving products and solutions, Midea actively promotes the integration of industrial knowledge.

(3) The stage of knowledge center

In the knowledge center phase, corporate universities are responsible for exploring and mastering industry frontier knowledge, engineering science knowledge and fostering

innovation incubation. As corporate universities continue to develop and improve, there is a growing emphasis on achieving a balanced ratio of internal and external teachers, thereby enhancing the relevance of teaching content delivered by enterprise teachers. The teaching content provided by all executives is mutually supportive and aligns with the development strategy of the enterprise. As knowledge management organizations, corporate universities collect the knowledge needs of various business departments, identify knowledge gaps of business departments and diagnose the knowledge barriers that impede various aspects of business processes. Moreover, at this stage, corporate universities also establish mechanisms to internalize external knowledge, share and apply imported management methods and theories within enterprises and organize employees to learn and internalize tacit knowledge. The promotion of knowledge ba not only improves the basic teaching facilities and learning environment but also cultivates mutual trust and teamwork consciousness between teachers and employees through cross-departmental communication, which makes the teaching auxiliary activities more effective.

With the evolution of corporate universities' functions, the intellectual capital network of corporate universities has become increasingly intricate, transforming into an intellectual capital network with the role of innovation bridge. The enhancement of the application incubation function of corporate universities depends on the comprehensive upgrading of teacher network capital, knowledge ba capital and knowledge process capital. The flow and integration of knowledge capital in corporate universities differ from the previous stage, requiring the connection of additional knowledge nodes, thereby further enhancing the interrelationships among nodes within the network. As a mature corporate university, Hisense College efficiently integrates three kinds of knowledge activities, extending its functions beyond knowledge transfer to actively undertaking knowledge innovation and aggregation.

According to an interviewee from Hisense College:

Hisense College established a technology research center in advance to assess the market and technical feasibility of new products and processes using evaluation procedures and technical methodologies.

The R&D department of the enterprise gains insights into market demand through the information platform established by Hisense College, which is then translated into research and development topics.

A dedicated project team was assembled for the research and development project, with engineers from both the production department and the R&D department at Hisense College working together to address technical challenges collaboratively.

An interviewee from Hisense College further told us:

Hisense College has actively contributed to the development of national and industry standards, thereby incorporating cutting-edge information and international technical certification standards into enterprise practices for development and design purposes.

5. Discussion, conclusion and implication

5.1 Discussion and conclusion

First, this paper explains the essence of intellectual capital and intellectual capital network of corporate universities. The study not only asserts that the intellectual capital of corporate universities comprises three dimensions, namely teacher network capital, knowledge process capital and knowledge ba, but also reveals that the interaction and network of knowledge among these three dimensions collectively construct the intellectual capital network of corporate universities. Intellectual capital is not merely a static intangible asset but rather a dynamic process of effectively harnessing knowledge (Ting *et al.*, 2020). Although existing literature generally holds the view that intellectual capital includes human capital, structural

capital and relational capital (Festa *et al.*, 2021; Chatterjee and Chaudhuri, 2022), limited attention has been given to exploring the specific dimension of intellectual capital that promotes the enhancement of knowledge management in corporate universities. According to the case study, it is observed that the categories of multiple intellectual capital in corporate universities align with the existing research on intellectual capital, but the composition and significance of specific dimensions of intellectual capital in corporate universities still differ from the existing literature. The human capital of a corporate university manifests in the knowledge, skills and experience accumulated by teachers, employees and enterprises, referred to as teacher network capital (Chen and Xu, 2022). The structural capital of a corporate university is represented by knowledge process capital. Nonaka *et al.* (2000) introduced the concept of knowledge ba, which pertains to the management of the context and conditions for knowledge transfer and creation. The relational capital of a corporate university symbolizes the knowledge ba. It is found that different intellectual capital elements in corporate universities can be considered as combinations of intellectual capital. This study explores how these three kinds of intellectual capital mutually reinforce each other and synergistically drive the development and enhancement of the knowledge management capability of corporate universities. Based on data coding, this paper constructs the model of intellectual capital combination of corporate universities, which is known as the “steering wheel” model of intellectual capital portfolio. Marinelli *et al.* (2023) incorporated intellectual capital into an ecosystem perspective and considered the social dimension of stakeholders. Although some scholars argue that the internal elements of intellectual capital and external capital are intricately intertwined (Yüksel *et al.*, 2022), and it also possesses multi-dimensional network attributes (Bontis, 1998; Inkpen and Tsang, 2005; Anderson and Parker, 2013; Abualqumboz *et al.*, 2021), existing research rarely studies the interconnection among various elements of intellectual capital from the perspective of knowledge network. This paper proposes that the corporate university, serving as an intermediary organization, connects the upstream and downstream industrial chains, external sources of knowledge, internal departments, employees and other nodes of enterprises. It establishes a linkage between intellectual capital and knowledge networks to form an intellectual capital network and manages the flow and transformation of internal and external intellectual capital of enterprises.

Second, this paper reveals three roles of the intellectual capital network in corporate universities, namely, coordinator, knowledge gatekeeper and innovation bridge. The majority of scholars believe that the primary role of intellectual capital is to improve enterprise performance (Subramaniam and Youndt, 2005; Lerro *et al.*, 2014; Shet *et al.*, 2021). Many scholars examine the role of intellectual capital by focusing on its measurement and reporting (Fiano *et al.*, 2020; Festa *et al.*, 2021). However, existing studies have not explored the effectiveness of enhancing organizational capacity and improving organizational performance from a network-based perspective of the role assumed by intellectual capital. Moreover, scholars have also proposed that corporate universities are open systems with permeability and cross-border characteristics (Patrucco *et al.*, 2017; Chen *et al.*, 2022), thus enabling the examination of the connections and roles of various elements of intellectual capital within corporate universities from a network perspective. Through an analysis of the characteristics of intellectual capital and knowledge networks of corporate universities, this paper finds that the intellectual capital network of a corporate university serves as the role of coordinator. It establishes a platform for education, training and knowledge exchange within enterprises by utilizing the intellectual capital network, which enables employees to actively engage in knowledge sharing. The knowledge gatekeeper organizes the external knowledge process of the enterprise, and then introduces the information, knowledge and skills acquired from the external knowledge network into the enterprise. The innovation bridge interacts with both the internal and external knowledge networks of enterprises, stimulating new ideas

and creativity among employees in the exploration of cutting-edge knowledge within the enterprise. Additionally, it provides systematic management and technical training for internal innovation incubation, while also leveraging external resources such as universities and research institutes to facilitate continuous technological development.

Third, this paper sheds light on the evolution of intellectual capital networks in corporate universities. Existing research contends that knowledge networks and intellectual capital are not static entities but rather dynamic in nature (Dumay and Garanina, 2013; Gabriele *et al.*, 2018; Marinelli *et al.*, 2023). Secundo *et al.* (2018b) propose distinct stages and characteristics in the development of intellectual capital. As corporate universities evolve, it is essential to adopt a dynamic perspective when examining the evolutionary process of intellectual capital within them (Chen *et al.*, 2022). While scholars generally agree that the activities undertaken by corporate universities are becoming increasingly diversified, and the demands for knowledge management functions of corporate universities are bound to dynamically evolve with the development of enterprises (Scarso *et al.*, 2017; Slettli and Grønhaug, 2017; Chen *et al.*, 2019). Knowledge management in corporate universities is regarded as an integrative process of intellectual capital and knowledge flow (Chen and Xu, 2022). Previous scholars put forward that the knowledge management activities of corporate universities evolve dynamically from knowledge upgrading and connection to innovation incubation (Petr and Elena, 2018; Chen *et al.*, 2022). Consequently, the knowledge management activities of corporate universities necessitate different requirements for intellectual capital and the supporting resources provided by the intellectual capital network. However, existing literature only explores the intellectual capital of corporate universities from a static perspective (Lerro *et al.*, 2014). To address this gap, it is crucial to investigate how intellectual capital and the intellectual capital network evolve and upgrade to effectively support knowledge management activities at different stages of corporate university development. In light of this, we propose a sequential inertial evolution model of the intellectual capital network in corporate universities. In essence, the endeavor to enhance the intellectual capital of corporate universities also entails an expansion process from teacher network capital and knowledge ba to internal and external knowledge processes. The three types of intellectual capital in corporate universities are coupled, and the internal and external network nodes gradually expand, assuming the role of an innovation bridge. In addition, through an analysis of the evolution of knowledge network-based intellectual capital and knowledge activities in corporate universities, this study identifies a synergistic relationship among intellectual capital, intellectual capital network and knowledge activities of corporate universities.

5.2 Theoretical implications

First, the existing literature appears to lack a comprehensive and adequate understanding of the intellectual capital and knowledge network of corporate universities. This paper enriches the theoretical research on knowledge management and intellectual capital in corporate universities. From the perspective of knowledge network, the paper reveals the compositional dimension of intellectual capital and the intellectual capital network of corporate universities. We discover a strong mutual reinforcing effect among teacher network capital, knowledge process capital and knowledge ba of corporate universities. The relationship among these three types of intellectual capital forms a “steering wheel” model. The intellectual capital and knowledge network of corporate universities interact to manage the internal and external flow of knowledge, thereby constituting the intellectual capital network of corporate universities. Second, the existing literature fails to emphasize the development path of intellectual capital of corporate universities. This paper explores the three roles undertaken by the intellectual capital network of corporate universities, namely, coordinator, knowledge gatekeeper and innovation bridge and further clarifies the mechanism of knowledge network-based intellectual capital of corporate universities. Additionally, this study reveals the dynamic characteristics of the intellectual

capital network and knowledge management in corporate universities. With the development of corporate universities, we observe a synergistic relationship among intellectual capital, the intellectual capital network and the knowledge activities of corporate universities. The evolution process of intellectual capital progresses from a singular form to richer and more expansive forms, forming a sequential inertia evolution model. Finally, our study enriches the literature on corporate universities. Previous studies primarily focused on corporate universities within a Western context, while this paper expands the research on corporate universities in the Chinese context.

5.3 Practice implications

First, corporate universities play a key role in knowledge management of enterprises. This paper aims to promote a heightened emphasis on the significance of knowledge management within corporate universities, with the ultimate goal of empowering organizations and employees and thereby enhancing their capabilities for knowledge innovation. To achieve this, we propose the enhancement of intellectual capital within corporate universities and the establishment of an interconnected intellectual capital network, aiming to effectively strengthen the knowledge management capabilities of corporate universities.

Second, the practical experiences of certain corporate universities summarized in this paper hold valuable insights for the advancement of corporate universities. For instance, it is essential for corporate universities to strike a balance between internal and external recruitment of teachers, establish network-based communication platforms to facilitate knowledge sharing, maintain close collaboration with various knowledge departments of enterprises and pay attention to integrating the knowledge process across business departments. These practices can provide a growth trajectory for other corporate universities. Drawing from the experiences of mature corporate universities, the evolution trend of intellectual capital highlights the importance of proactive accumulation of industry-leading knowledge and organizing business process knowledge within enterprises.

Moreover, it is crucial for corporate universities to construct internal and external knowledge connection platform, along with an innovation incubation center, forming an intellectual capital network that serves as an innovation bridge. This will effectively promote the advancement of knowledge management in corporate universities. To enhance knowledge management capabilities, corporate universities should engage with internal and external stakeholders of enterprises. Internally, corporate universities can strengthen knowledge exchange and collaboration among various departments through employee training programs, knowledge-sharing platforms and seminars.

Furthermore, corporate universities should identify the knowledge needs and gaps in production, research and development and other business departments. Externally, they should selectively gather relevant external information and integrate introduced external knowledge and resources. Additionally, replicating research collaborations with universities or research institutes can be effectively applied to other corporate universities.

Finally, this paper provides guidance to corporate universities in establishing a knowledge management platform and enhancing their ability to connect internal and external knowledge resources, thereby fostering new knowledge creation. It offers practical suggestions to position corporate universities as premier learning organizations and catalysts for enterprise innovation. The guidance is valuable for corporate universities seeking to enhance their capacity to provide knowledge management services for enterprises.

5.4 Limitations and future research

First, in order to enhance the memorability and comprehensibility of the analysis results, it is often necessary to appropriately simplify the evolution process and various characteristics of

the research subjects. However, it is important to acknowledge that the selected corporate university cases in this study possess certain unique attributes, which may have an impact on the reliability of the study to some extent. Follow-up research can enrich the case studies of corporate universities and address the aforementioned research limitations.

Second, due to the limited development of the theory regarding intellectual capital and its evolutionary mechanism of corporate universities, this paper adopts the case study method. However, it is worth noting that incorporating non-metric factors can be challenging, potentially resulting in missing indicators and insufficient precision. In the future, quantitative research methods can be utilized to validate the findings of this paper.

Finally, the intellectual capital of corporate universities has not been measured in this paper. Measurement can offer a better understanding of how intellectual capital is transformed into value that is beneficial for enterprises. To effectively manage and enhance corporate universities, future studies can adopt measurement methods to investigate and report on the intellectual capital of corporate universities.

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